

Arnibish Ray

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EDUCATION

Northeastern University, Boston, MA

GPA: 3.95

B.S. in Mechanical and Bioengineering | Minor: Business Administration

Expected April 2028

Honors: John Martinson Honors Program | Dean's List (All Semesters) | Peak Award Fellowship (F25, S26) | Senior Standing

Relevant Coursework (Lab*): ME Design, Fluid Mechanics*, Circuits*, Thermodynamics, Materials Science*, Dynamics*

Activities: Residential Assistant | Northeastern Running Club (E-Board) | Forge (E-Board) | ASME (Instructor)

Global Experience: Dialogue of Civilizations, Greece, Evolution of Scientific Thought & Technology

SKILLS

Design & Simulation: SolidWorks (Advanced, CSWP Certified) | AutoCAD | GD&T

Fabrication & Testing: 3D Printing | Laser Cutting | Mechanical Assembly | Electromechanical Testing | Sensors & Actuators | DFM

Programming & Analysis: MATLAB | C++ | Data Analysis | Excel | Microsoft Office

CERTIFICATIONS

Certified SolidWorks Professional (CSWP) | Excel Essential Training | MATLAB Essential Training | OSHA / PPE Safety Training

EXPERIENCE

Mechanical Systems Engineer Co-op | Nova Biomedical, Waltham, MA

Jan 2026 - Present

- Designed and fabricated a SolidWorks-based sheet metal test fixture functioning as an independent power supply to validate 4 CDV modules (Osmo, ESM, RSM, IGG) outside the Flex2 analyzer.
- Performed system validation across cell analyzers and CDV machines, including display verification, dilution accuracy, RFID board integration, and ML-based cell counting performance.
- Prepared and concentrated CHO/HEK cell samples for validation testing, then analyzed results in Excel and compiled reports to identify failure trends and drive targeted fixes.
- Maintained and revised test protocols, SOPs, and BOMs, and fabricated/updated fixture parts for iterative design changes.

Biomechanics Research Assistant | Northeastern University LabSYNC, Boston, MA

Oct 2024 - Present

- Built a MATLAB-based data pipeline processing IMU data from 144 injured and non-injured runners, extracting kinematic features and applying a classification model to assess injury risk with 75% accuracy; presented findings at Northeastern RISE Expo.
- Implemented a CARL (Continuous Activity Recognition Logistic) model to segment running vs. non-running intervals from raw accelerometer data, isolating gait-specific features for downstream analysis.
- Analyzed hip MRI data from 30 adolescent athletes to study femoral morphology's role in load distribution and injury risk.

LEADERSHIP & VOLUNTEERING

Organizational Developer | Forge Product Development Studio, Boston, MA

Jan 2026 - Present

- Organized a co-op industry panel, including company outreach, event coordination, and student engagement, for 50 attendees.
- Managed budgeting, purchasing, and resource allocation for student engineering workshops and project showcases.

Treasurer & Captain | Northeastern Club Running Team, Boston, MA

Jan 2026 - Present

- Manage budgeting, funding allocation, and fundraising for a 200+ member organization.
- Coordinate sponsorship outreach and financial planning for races, travel, and club events.

FIFA World Cup 2026 Volunteer | Boston Stadium, Boston, MA

Jun 2026 - Jul 2026

- Concessions worker, supporting stadium operations, acted as an information guide, and assisted fans with wayfinding.

PROJECTS

Gesture-Controlled 6-Axis Robotic Arm | PEAK Fellowship Project

May 2026 - Present

- Designed and assembled a 3D-printed 6-axis robotic arm and gripper in SolidWorks with functional joints, linkages, and motion.
- Integrated an Arduino-based wireless glove controller using IMU sensors to map hand motion to servo actuation.
- Calibrated gripper motion and validated grasp repeatability, achieving an 80% success rate across 30 trials.

Live-Vital | Independent Project

Jun 2025 - Jul 2025

- Designed a MAX30100 biosensor system integrating PPG signal acquisition, embedded systems, and real-time HR/SpO2 feedback, housed in a 3-part SolidWorks enclosure optimized for durability and manufacturability.
- Built Arduino-based feedback mechanisms (LCD display, speaker, LED indicators) for real-time user output.
- Validated system performance against smartwatch data, achieving 96% accuracy.

INTERESTS

Distance Track & Field Running | Weightlifting | Soccer | DIY Engineering | 3D Printing